

CS-I300B

Auto-Chemistry Analyzer



Specification

System Type	Random access,fully automatic, discrete, STAT priority
Throughput	Constant speed 900T/H for colorimetric test, 1260T/H for ISE
Light Source	Long life halogen lamp
Wavelength	340~800nm, 12 wavelengths
Analysis Method	End-point, kinetics, fixed-time, etc.
Calibration Method	1 point method, 2 point method, multiple point linear method, non-linear method
Probe	Independent sample probes, reagent probes, automatic liquid level detection, probe liquid level tracing and clot detection function
Rinsing Mechanism	Probe inner wall high pressure rinsing
Mixing Mechanism	2 independent mixers ensure sufficient reaction
Reaction Cuvette	160 new type reaction cuvettes
Temperature Control	The temperature of reaction disk incubation bath is 37℃±0.1℃
Sample Tube	140 sample positions, supporting multiple tubes and sample cups
Reagent Position	2 compartments, 132 reagent positions totally, with continuous refrigeration; supporting 4 kinds of reagent test
Sample Volume	1.5 ul~35ul, 0.1ul increment
Reagent Volume	15 ul~350ul, 1ul increment
Reaction Volume	Minimum reaction volume 100μl~450μl



Certified to
ISO 9001:2008 and ISO 13485:2003

DIRUI INDUSTRIAL CO.,LTD.

3333 Yiju Street,New&High Tech. Development Zone
Changchun,Jilin 130103,P.R.China
Tel: +86(431)81935329 85100409
Fax: +86(431)85172581 85083741
E-mail: dirui@dirui.com.cn Http://www.dirui.com.cn
• Specifications subject to change without notice.

20160612

DIRUI



CS-I300B

Auto-Chemistry Analyzer



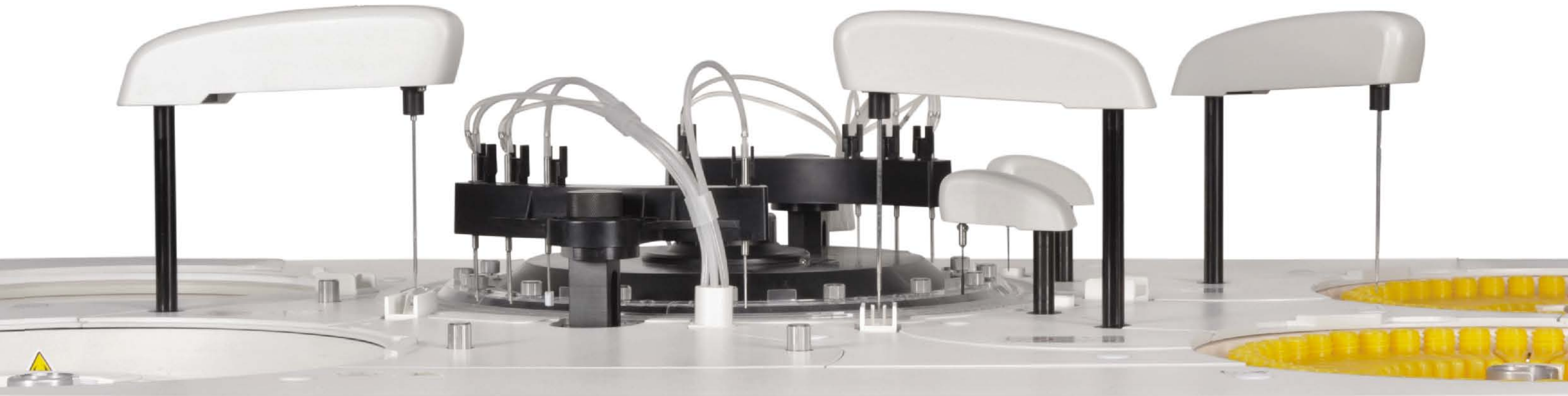
DIRUI

The Next Step in Clinical Chemistry Evolution

DIRUI CS series auto-chemistry Analyzers have been installed in over 30 countries world wide. Through our dedication to quality manufacturing and world class customer service, Dirui has established itself as one of the world's foremost clinical chemistry analyzer manufacturers.

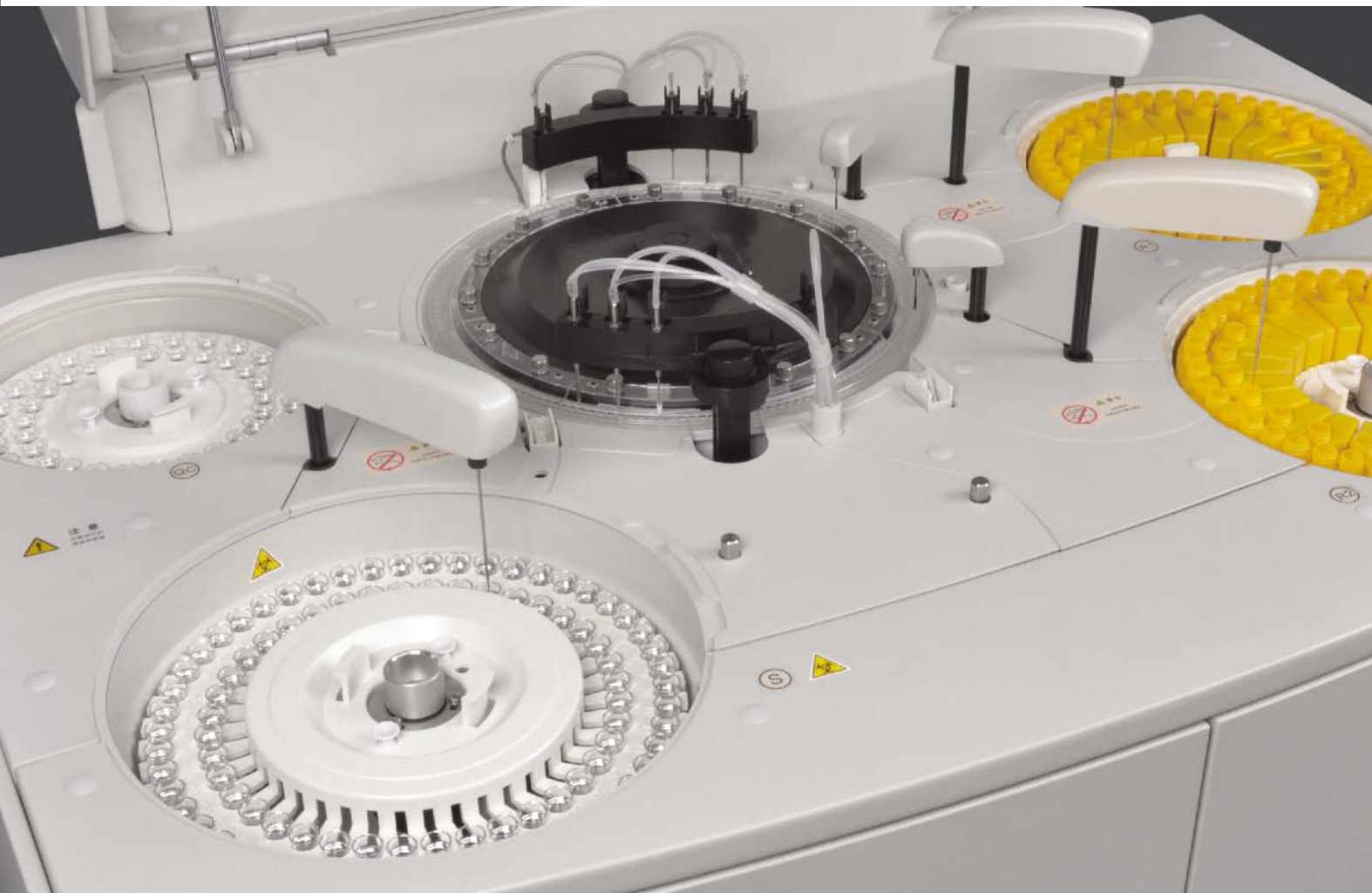
CS-I300B

Auto-Chemistry Analyzer



CS-I300B

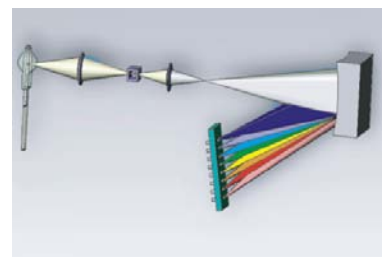
Auto-Chemistry Analyzer



Advanced Photometry System

Holographic concave flat field grating, rear spectrophotometry

- Cluster-condensing light(point light source) technology to realize microvolume analysis, less reagent consumption
- Long life light source adopts circulating water cooling method
- Light source with best position design, no signal attenuation, strong anti-interference
- Anti-ambient light interference to get accurate result



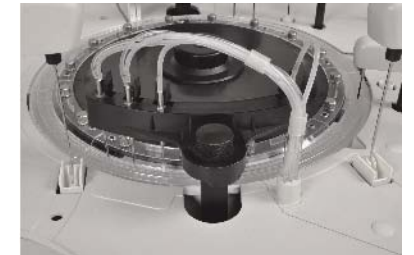
Stable Temperature Control System

A. Cooling System

- Advanced postposition semi-conductor helps direct heat releasing, water cooling to ensure stable temperature and easy maintenance

B. Constant Temperature Device of Reaction Cuvette

- Recycling water of constant temperature, automatically changing water and adding defoamer. Reaction cuvettes are immersed into warm water which heats the cuvettes evenly and reduces ambient temperature influence, no need of reagent pre-heating
- PID thermostat technology ensures 37°C, $\pm 0.1^\circ\text{C}$ variation of temperature control



Accurate Sample/Reagent Pipetting Mechanism

Probe:

- Polished probes with nano processing technology, reducing the cross-contamination effectively
- Automatic liquid level detection: the probes automatically detect the liquid level and stop descending ensuring the probes enter into liquid at perfect depth and reducing liquid suspension
- Collision detection function, self-resetting, automatic sample and reagent pipetting
- Intelligent clog and clot detection: detecting the status of probe clog and the existence of clot
- High pressure rinsing function, enhancing pipetting volume accuracy. High pressure rinsing for inner wall of 3 probes, water fall rinsing for outer wall



Syringe:

- Long life high-precision ceramic piston ensures high precision of sampling, low maintenance cost
- Water degassing technology: The analyzer has special degassing device to remove the air dissolved in the tube system, which ensures quick, accurate and microvolume pipetting.
- The analyzer senses the remaining volume of samples and reagent and the number of test items automatically and alarms when the samples and reagent are not enough to ensure continuous analysis.



High-Efficient Rinsing System

A. Automatic Cuvette Rinsing

- 8 stops 12 steps rinsing by warm water ensures complete cleaning

B. Inner Wall Rinsing of probe

- Draining liquid by vacuum, detergent and warm water high pressure rinsing
- Carryover contamination rate $\leq 0.1\%$, ensures best cleaning status



CS-1300B

Auto-Chemistry Analyzer

Calibration and QC Function

- Linear and non linear calibration;With manual and auomatic calibration function availabe to select
- 9 types of calibration curve fitting formulas to satisfy different item analysis requirement
- Each item can use 6 different levels' calibrator at most
- With calibration tracing function;Change-trend chart of K value helps to reduce system error
- Monthly QC ensure that the instrument is working at the best analysis condition
- QC rule: default as Westgard multi rules
- QC chart: depict and print the relevant L-J chart , Cumulative automatically
- Out of control reason and original record function, which complies with the lab QC management

Operating Software

- English version graphical operating software, user friendly operation interface
- Real-time online help system, reduces downtime
- Easy software operation, advanced function, consistent with the use requirement of clinical test
- Multiple self-monitoring functions ensure the intelligentialize of analysis process
- Data dictionary and information input, which is convenient and efficient
- Multiple data query, statistic and report printing function
- Same patient information will be memorized by the software automatically and input automatically, all the things will be completed by one button
- Multiple report formats, user define modification is supported

Monitoring and Calibration Function

- Linear limit over, reference limit over, substrate depletion, antigen surplus, no reaction equibrum point checking function
- Serum checking function avoids interference from hemolytic, lipemic, icteric reagent
- Against cross-contamination program, avoid interference from different reagents, samples and reaction cuvettes
- Instrument analysis alarm hierarchical handling to arrange the job sequence by priorities
- User permission hierarchical management to enhance system security



Barcode Scanning

- Automatic routine sample barcode scanning, accurate sample information recognition
- Automatic reagent barcode scanning, accurate, fast reagent information acquisition

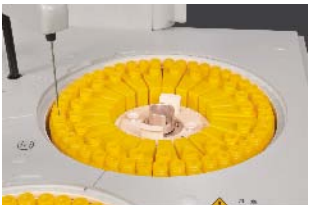
ISE Module(Optional)

- Indirect simultaneous measurement of K^+ , Na^+ , Cl^-
- Throughput of 450 T/H
- Long life electrodes



User-friendly Design

- **Reagent Disks:**
- Newly designed slope angle of reagent disk ;Large capacity of reagent postions
- Two disks allow 132 items to be tested at most
- Flexible Sample Positions:sample cups and original tubes with the diameter of 12-16mm are available



Advanced Analysis System

- High precision instrument, high quality reagents, Control and Calibrator with complete traceability system contribute to advanced analysis system

